21ASC305T
ROCKET PROPULSION

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- ***Unit-1 - Basic Concepts 9 Hour***
- *Rocket propulsion systems -working principle, Classification of rockets, Rocket equation, Rocket Nozzles and its classifications -Nozzle performance- equilibrium and frozen flow, Characteristic velocity and Thrust coefficient, Performance parameters and efficiencies of rocket, Staging and clustering of rocket*
- ***Unit-2 - Chemical Propellants 9 Hour***
- *Chemical propellants -Molecular mass, Specific heat capacities, Specific heat ratio, mixture ratio, Stoichiometric ratio and equivalence ratio, Energy release during combustion-Heat of formation and Heat of combustion-Combustion Instabilities-Criterion for choice of propellant, Solid Propellants-Composition and processing, Liquid Propellants-Classifications of liquid propellant, Storability of liquid propellant, Chemical Equilibrium*
- ***Unit-3 - Solid Rocket Engine 9 Hour***
- *Propulsion Elements for Solid Rocket Motors, Solid propellant combustion and Internal Ballistics of Motors - Mechanism of burning, Ignition System-Igniter types - Solid propellant grain design - Burn Rate-Factors influencing burn rates, burn rate index for stable operation, Action time and burn time, Design of Solid Propellant rocket, Simulation of solid rocket motor performance.*
- ***Unit-4 - Liquid Rocket Engine 9 Hour***
- *Liquid Propellant Rocket-Hardware components and its functions, Propellant feed systems –Gas pressure feed system, Turbo pump feed systems and engine cycles, Tank Pressurization - Types of injectors – Thrust chamber, Cooling of Thrust chamber - Cryogenic propulsion system and its special features, Simulation of Liquid rocket performance, Hybrid rockets- Review of solid-fuel regression rate behavior in classical and non-classical Hybrid Rocket Motors - Rocket engine design using software.*
- ***Unit-5 - Non-conventional Propulsion Techniques 9 Hour***
- *Non - conventional propulsion techniques -Electrical Rockets-Electro-thermal propulsion system-Arc-jet thruster, Resistojet Thruster-Electro-static thruster-Ion thruster, Bombardment Ionization, Hall Effect Thruster -Electro-magnetic propulsion system-Magneto plasma dynamic accelerator, Pulsed Plasma accelerator- Laser Propulsion -Nuclear powered rockets- Solar Propulsion system*

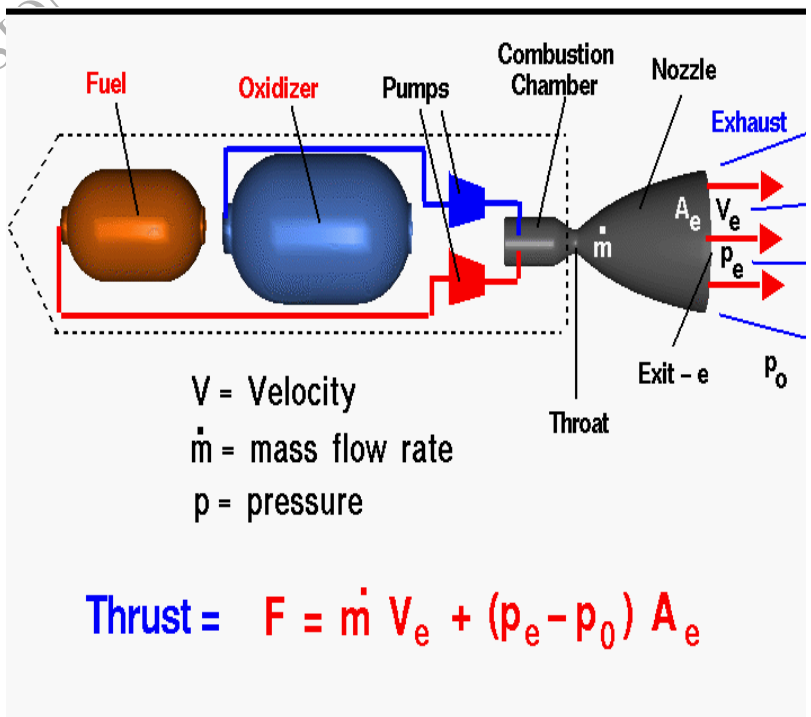
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- *1. George P. Sutton, Oscar Biblarz, "Rocket propulsion elements", Wiley India Pvt Ltd. Ninth Edition, 2017*
- *2. Ramamurthi.K, "Rocket propulsion", Laxmi Publications, India, Second edition, 2019.*
- *3. J D. Mattingly and K M. Boyer,"Elements of Propulsion: Gas Turbines and Rockets", Second Edition,2016*
- *4. Philip Hill and Carl Peterson, "Mechanics and thermodynamics of propulsion", Pearson India, second edition, 2010.*
- *5. Stephen R. Turns, "An Introduction to Combustion: Concepts", McGraw-Hill Education; 4th edition, 2020*
- *6. Yahya S M, "Gas Tables",8th Edition, 2018*

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Liquid propellant rocket-types

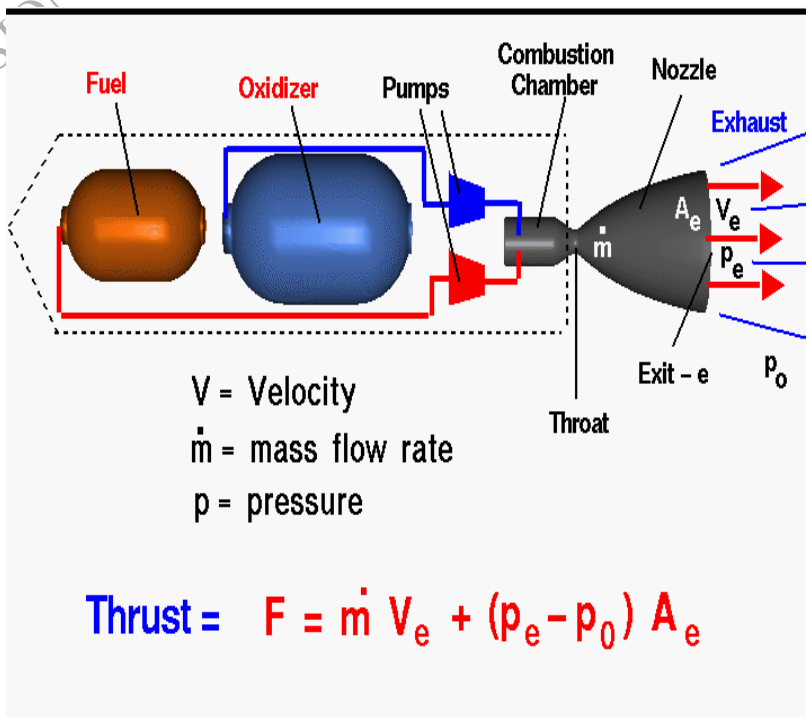
- The *rocket engine* consists of
  - ***Thrust chambers***
  - a ***feed mechanism*** for supplying the propellants from their tanks to the thrust chamber(s),
  - a ***power source*** to furnish the energy for the feed mechanism,



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Liquid propellant rocket-types

- suitable ***plumbing*** **or** ***piping*** to transfer the liquid propellants under pressure,
- A ***structure*** to transmit the thrust force, and
- ***control devices*** (including valves) to start and stop and sometimes also to vary the propellant flow and thus the thrust.

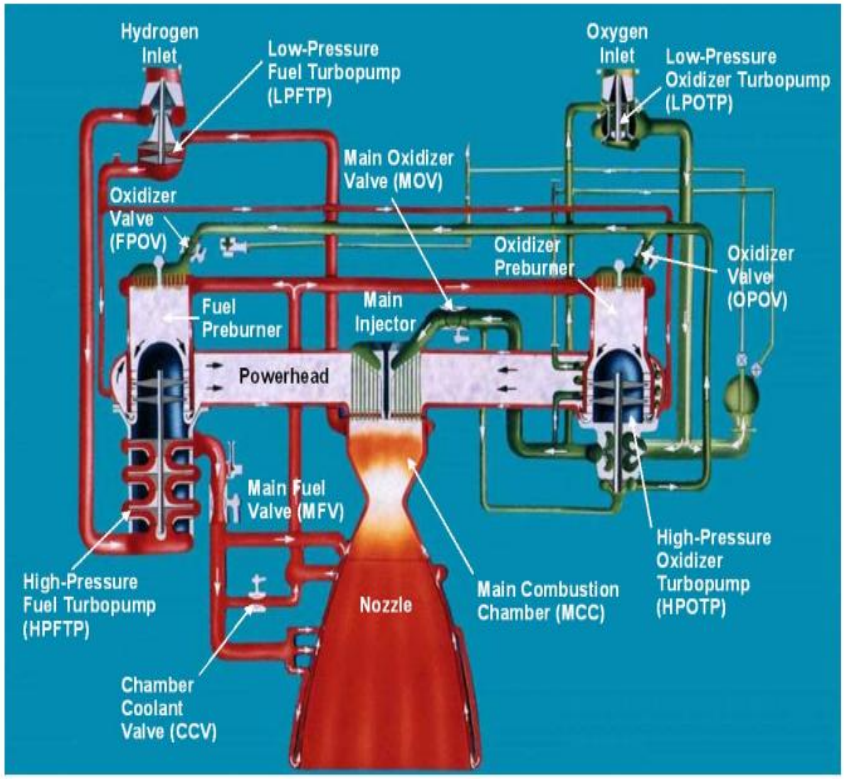


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Liquid propellant rocket-types

- The propellant control system or valves and actuation systems in a rocket engine, often collectively governed by the **thrust control system or Engine Control Unit (ECU).**

- The **Engine Control Unit (ECU)** or flight computer constantly calculates the required thrust based on the rocket's current speed, altitude, weight, and trajectory.



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Liquid propellant rocket-types

- **Propellant Control Valves (Main Valves):** These are the physical mechanisms that open or close to start and stop the flow of fuel and oxidizer.

- **Throttling Valves / Cavitating Venturis:** To vary the flow rate (and thus change the amount of thrust), engines use variable-geometry valves or adjustable flow controllers. This process is called **thrust throttling**.

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Liquid propellant rocket-types

- **Actuators:** These are the mechanical, hydraulic, or electrical drivers that physically move the valves to the precise positions commanded by the flight computer.

- **Pressure sensors and flow meters** inside the engine constantly monitor the actual propellant flow. If the thrust is too low or too high, the computer detects the error in milliseconds.

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Liquid propellant rocket-types

- Propellants, the working substances of rocket engines, constitute the fluid that undergoes chemical and thermodynamic changes.

- The term ***liquid propellant*** embraces all the various propellants stored as liquids and may be one of the following

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Liquid propellant rocket-types

- 1. Oxidizer (liquid oxygen, nitric acid, nitrogen tetroxide, etc.).
- 2. Fuel (kerosene, alcohol, liquid hydrogen, etc.).
- 3. Chemical compound (or mixtures of oxidizer and fuel ingredients) capable of self-decomposition, such as hydrazine (can chemically break down by itself and release energy, without needing a separate external fuel or oxidizer.)
- 4. Any of the above, but with a gelling agent (these have yet to be approved for production). Gelled kerosene, Gelled LOX, Gelled hydrazine

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Typical Characteristics of Two Categories of Liquid Propellant Rocket Engines

Purpose/Feature	Boost Propulsion	Auxiliary Propulsion
Mission	Impart significant velocity to propel a vehicle along its flight path	Attitude control, minor space maneuvers, trajectory corrections, orbit maintenance
Applications	Booster stage and upper stages of launch vehicles, large missiles	Spacecraft, satellites, top stage of antiballistic missile, space rendezvous
Total impulse	High	Low
Number of thrust chambers per engine	Usually 1; sometimes 4, 3, or 2	Between 4 and 24
Thrust level per thrust chamber	High; 4500 N up to 7,900,000 N or 1000–1,770,000 lbf	Small; 0.001 up to 4500 N, a few go up to 1000 lbf
Feed system, typical	Mostly turbopump type; occasionally pressurized feed system for smaller thrusts	Pressurized feed system with high-pressure gas supply
Tank pressure range	0.138–0.379 MPa or 20–55 psi	0.689–17.23 MPa or 100–2500 psi
Most common cooling method	Propellant cooled	Radiation cooled
Propellants (see next section)	Cryogenic and storable liquids	Storable liquids, monopropellants, and/or stored cold gas
Chamber pressure	2.4–21 MPa or 350–3600 psi	0.14–2.1 MPa or 20–400 psi
Number of starts during a single mission	Usually no restart; sometimes one, but up to four in some cases	Several thousand starts for some space missions
Cumulative duration of firing	Up to a few minutes	Up to several hours
Shortest firing duration	Typically 5–40 sec	0.02 sec typical for pulsing small thrusters
Time elapsed to reach full thrust	Up to several seconds	Usually very fast, 0.004–0.080 sec
Life in space	Hours, days, or months	Up to 15 years or more in space

LIQUID PROPELLANTS

- A ***monopropellant*** may contain an oxidizing agent and combustible matter in a single liquid substance.

- It may be a stored mixture of several compounds or it may be a homogeneous material, such as hydrogen peroxide or hydrazine.

- Monopropellants are stable at ambient storage conditions but decompose and yield hot combustion gases when heated or catalyzed in a chamber.

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LIQUID PROPELLANTS

- A ***bipropellant*** consists of two separate liquid propellants, an oxidizer and a fuel.
- They are stored separately and are mixed inside the combustion chamber.
- A ***hypergolic*** bipropellant combination *self-ignites* upon contact between the oxidizer and the liquid fuel.
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- A ***nonhypergolic*** bipropellant combination needs energy to start combusting (e.g., heat from an electric discharge) and such engines need an ignition system.

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LIQUID PROPELLANTS

- A ***cold gas propellant*** (e.g., helium, argon, or gaseous nitrogen) is stored at ambient temperatures but at relatively high pressures; it gives a comparatively low performance but allows a simple system, and is usually very reliable.

- They have been used for roll control and attitude control.

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LIQUID PROPELLANTS

- **Satellite Attitude Control:** Managing the precise pointing, orientation, and alignment of satellites in orbit.
- **Station Keeping:** Small adjustments to maintain a satellite's specific orbital slot or position.
- **CubeSats and Nano-Satellites:** Providing primary or secondary propulsion for very small spacecraft where system weight and volume budgets are tight.
- **Spacecraft Reaction Control Systems (RCS):** Handling fine translational and rotational movements during docking or tight orbital maneuvers.
- **Manned Spaceflight Maneuvering Units:** Historical examples include the Handheld Maneuvering Unit (HHMU) and the Manned Maneuvering Unit (MMU) used by NASA astronauts for spacewalks.
- **Launch Vehicle Pressurization Support:** Acting as a high-pressure pressurant gas (especially helium) to push liquid propellants into rocket combustion chambers

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LIQUID PROPELLANTS

- A ***cryogenic propellant*** is a liquefied gas at lower than ambient temperatures, such as liquid oxygen (−183 ∘C) or liquid hydrogen (−253 ∘C).

- Provisions for venting the storage tank and minimizing vaporization losses are necessary with this type.

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LIQUID PROPELLANTS

- ***Storable propellants*** (e.g., nitric acid or gasoline) are liquid at ambient temperatures and at modest pressures and can be stored for long periods in sealed tanks.

- ***Space-storable propellants*** remain liquid in the space environment; their storability depends on the
  - specific tank design,
  - thermal conditions, and
  - tank pressures.

- An example is ammonia.

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LIQUID PROPELLANTS

- A *gelled propellant* is a thixotropic liquid with a gelling additive.

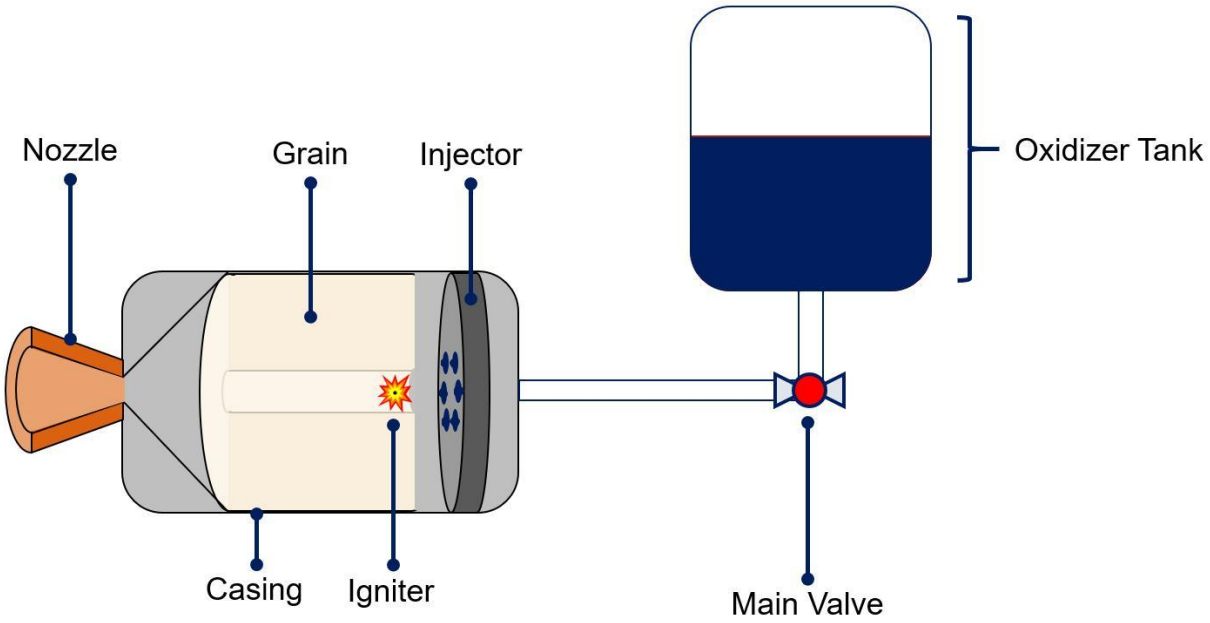
- It behaves in storage as a jelly or thick paint (it will not spill or leak readily) but can flow under pressure and will burn, thus being safer in some respects.

- Gelled propellants have been used in a few experimental rocket engines but, to date, gelled propellants have not been in production.

- *Hybrid propellants* usually have a liquid oxidizer and a solid fuel.

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What is a Hybrid Rocket Engine?



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LIQUID PROPELLANTS - PROPELLANT PROPERTIES

- It is important to distinguish between characteristics and properties of *liquid propellants* (i.e., fuel and oxidizer liquids in their unreacted condition) and those of the *hot gas mixture* resulting from their reaction in the combustion chamber.

- The **chemical nature of liquid propellants** and **their mixture ratio** determine the properties and characteristics of both storage and reaction products.

- Because none of the known practical propellants encompass all properties deemed desirable, the selection of propellant combinations is usually a compromise between various economic factors,

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LIQUID PROPELLANTS

- **Economic Factors**
- *Availability* in quantity and a *low cost* are very important considerations in propellant selection.

- In military applications, consideration has to be given to the *logistics* of production, supply, storage, along with other factors.

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LIQUID PROPELLANTS

- **Economic Factors**
- The production process should require only ordinarily available chemical equipment and available raw materials.

- It is more expensive to use toxic or cryogenic propellants than storable, nontoxic ones, because the former require additional steps in their
  - operation,
  - more safety provisions,
  - additional design features,
  - longer check-out procedures prior to launch, and
  - often better trained personnel.

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LIQUID PROPELLANTS

- **Performance of Propellants**
- The performance rocket engines may be compared on the basis of their *specific impulse*, *exhaust velocity*, *characteristic velocity*, and/or other engine parameters.

- The **specific impulse and exhaust mixture ratio**, **velocity** are functions of
  - pressure ratio,
  - specific heat ratio,
  - combustion temperature,
  - mixture ratio and
  - molecular mass .

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- **Performance of Propellants**

- Very often, performance is also expressed in terms of *flight performance parameters* for specific rocket applications.

- Average propellant density, total impulse, and engine mass ratio usually enter into the various flight relation descriptions.

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- For **high performance**, *high chemical energy content* per unit of propellant mixture is desirable because this yields high chamber temperatures.

- A ***low molecular mass*** for the combustion products gases is also desirable.

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LIQUID PROPELLANTS

- This can be accomplished by using fuels rich in hydrogen obtained when a significant portion of the hydrogen gas injected or produced remains uncombined.

- In general, therefore, the best mixture ratio for many bipropellants is not stoichiometric (which results in complete oxidation and yields the highest flame temperature) but fuel-rich, containing a large amounts of low-molecular-mass reaction products.

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$$V_e \propto \sqrt{\frac{T_c}{M}}$$

LIQUID PROPELLANTS

- ***Gelled propellants*** are materials that have additives that make them thixotropic.

- The **additives** in gelled propellants are usually called **gelling agents** or **rheological modifiers**. They are added to a liquid fuel or oxidizer to make it behave like a **gel**.
- **Common types of gelling agents**
  - For **gelled fuels**, examples include:
  - **Organometallic compounds**
  - **Polymeric thickeners**
  - **Silica-based materials**
  - Certain **metal soaps** and surfactant-type additives

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LIQUID PROPELLANTS

- For example, a hydrocarbon fuel can be thickened using a suitable **polymeric or particulate gelling agent**.

- They have the consistency of thick paint or jelly when at rest, but do liquefy and flow through pipes, valves, pumps and/or injectors when an **adequate pressure or shear stress is applied.**

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LIQUID PROPELLANTS

- **Desirable Physical Properties**
- ***Low Freezing Point.*** This permits operation of rockets in cold environments.

- The **addition of small amounts of freezing point depressants** has been found to help lower the freezing point in some liquid propellants that might otherwise solidify at environmental storage conditions.

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LIQUID PROPELLANTS

- ***High Specific Gravity.*** Denser propellants provide a larger propellant mass for a given vehicle tank volume.
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- Alternatively, **for a given mass**, they permit smaller tank volumes and, consequently, lower structural vehicle mass and lower aerodynamic drag.

- The specific gravity of a propellant, therefore, has an important effect on the maximum flight velocity and range of any rocket-powered vehicle or missile flying within the Earth's atmosphere.

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LIQUID PROPELLANTS

- ***Stability.*** Proper *chemical stability* means no decomposition of the liquid propellant during operation or storage, even at elevated temperatures.

- With many propellants, insignificant amounts of *deterioration* and/or *decomposition* during long-term (over 15 years) storage and minimal *reaction with the atmosphere* have been attained.

- **A desirable liquid propellant** should also experience no *chemical deterioration* when in contact with tubing, pipes, tank walls, valve seats, and gasket materials even at relatively high ambient temperatures.

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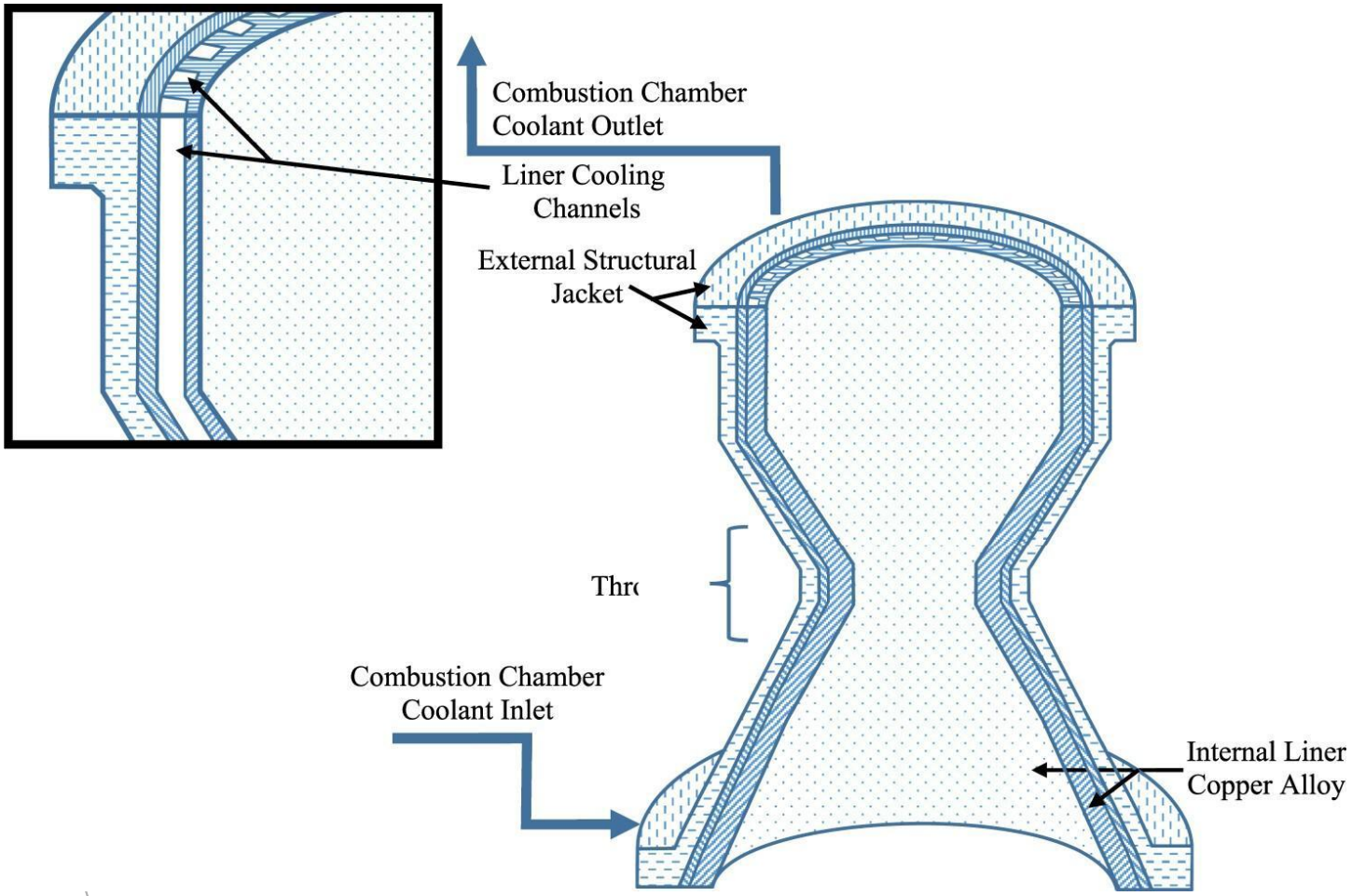
LIQUID PROPELLANTS

- **No** appreciable ***absorption of moisture*** and no adverse effects of small amounts of ***impurities*** are also desirable properties.

- There should also be **no appreciable chemical deterioration** when liquids flow through hot cooling jacket passages of a regeneratively cooled thrust chamber.

- When carbon-containing coolants decompose and form (carbonaceous) deposits on hot inside surfaces of cooling passages, such deposits may
  - harden,
  - reducing the heat flow,
  - increasing metal temperatures locally, and
  - thus may cause the unit to weaken and eventually fail.

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Combustion Chamber
Coolant Outlet
Liner Cooling
Channels
External Structural
Jacket
Thr
Combustion Chamber
Coolant Inlet
Internal Liner
Copper Alloy

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LIQUID PROPELLANTS

- ***Heat Transfer Properties.***
  - *High specific heat, (*absorb a relatively large amount of heat for a given temperature rise)
    - $Q=mc_p\Delta T$
  - *high thermal conductivity*,
  - *low freezing temperature*, and
  - *high boiling or decomposition temperature*

are all desirable properties for propellants used for thrust chamber cooling

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Storability

- Some **pressure feed systems** can be prefilled with storable propellant and pressurizing agent at the factory and stored in readiness for operation.

- **Compared to solid propellant rocket units**, these prepackaged liquid propellant pressurized feed systems offer advantages in
  - long-term storability,
  - random restarts, and
  - better resistance to transportation vibration or shock.

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Chemical equilibrium

- ***Chemical equilibrium*** occurs in reversible chemical reactions when the rate of product formation exactly equals the reverse reaction (one forming reactants from products).

- Once this equilibrium is reached, no further changes in concentration take place.

- All three gases would be present in relative proportions that would depend on the pressure, temperature, and equilibrium state of the mixture

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- The *combustion process* normally occurs at essentially constant pressure (isobaric) in the combustion chamber and the resulting gases follow Dalton's law.

- Chemical reactions do occur very rapidly during propellant combustion.

- The chamber volume is assumed to be large enough and the residence time long enough for attaining chemical equilibrium within the chamber.

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